SN Ia – BAO Locking & De-chirp Analysis (v0V1)

Executive summary

We analyze Type Ia supernova (SN Ia) Hubble-diagram residuals $\Delta\mu$ on a uniform $L=\log(1+z)$ grid with the Hilbert–Huang Transform (HHT; CEEMDAN). By constructing a de-chirped coordinate $U(L)$ from a BAO proxy $b\alpha = (1-\alpha)\cdot(DM/rd) + \alpha\cdot(DH/rd)$, we find that $\alpha \approx 0.375$ minimizes the scatter (MAD) of IMF2–3 instantaneous frequencies when re-analyzed in U. For this configuration, IMF2 and IMF3 show MAD_U/MAD_L ≈ 0.57 and 0.44, respectively, consistent with the bar-chart tightening across IMFs. Surrogate tests yield $p \approx 10^{-4}$ for the focus-IMF energy statistic. A window-width (WL) sweep of the locking metric shows increasingly negative scores for IMF2–3 with larger WL; by a simple |IMF2,IMF3| figure-of-merit, WL ≈ 0.05 performs best.

Acronyms & symbols

HHT: Hilbert–Huang Transform. CEEMDAN: Complete Ensemble Empirical Mode Decomposition with Adaptive Noise. IMF: Intrinsic Mode Function. L: log(1+z). U(L): de-chirped coordinate derived from a BAO proxy. BAO: Baryon Acoustic Oscillation. DM/rd, DH/rd: transverse and radial BAO-scale proxies normalized by the sound horizon rd. MAD: median absolute deviation. WL: sliding window width in L used in the locking score.

Key definitions

Instantaneous frequency (per sample) for IMF i: $f\_i(L) = (1/2\pi) \cdot d/dL \arg(\text{Hilbert}(IMF\_i(L)))$.

BAO mix: $b\_\alpha(L) = (1-\alpha)\cdot(DM/rd)(L) + \alpha\cdot(DH/rd)(L)$.

De-chirp map: $U(L) = CDF(|b\_\alpha(L)|) = (\int|b\_\alpha(L')|\, dL') / (\int|b\_\alpha(L')|\, dL')\_range \in [0,1]$.

Tightening per IMF: $100\times(1 - MAD\_U/MAD\_L)$ [%].

Core figures

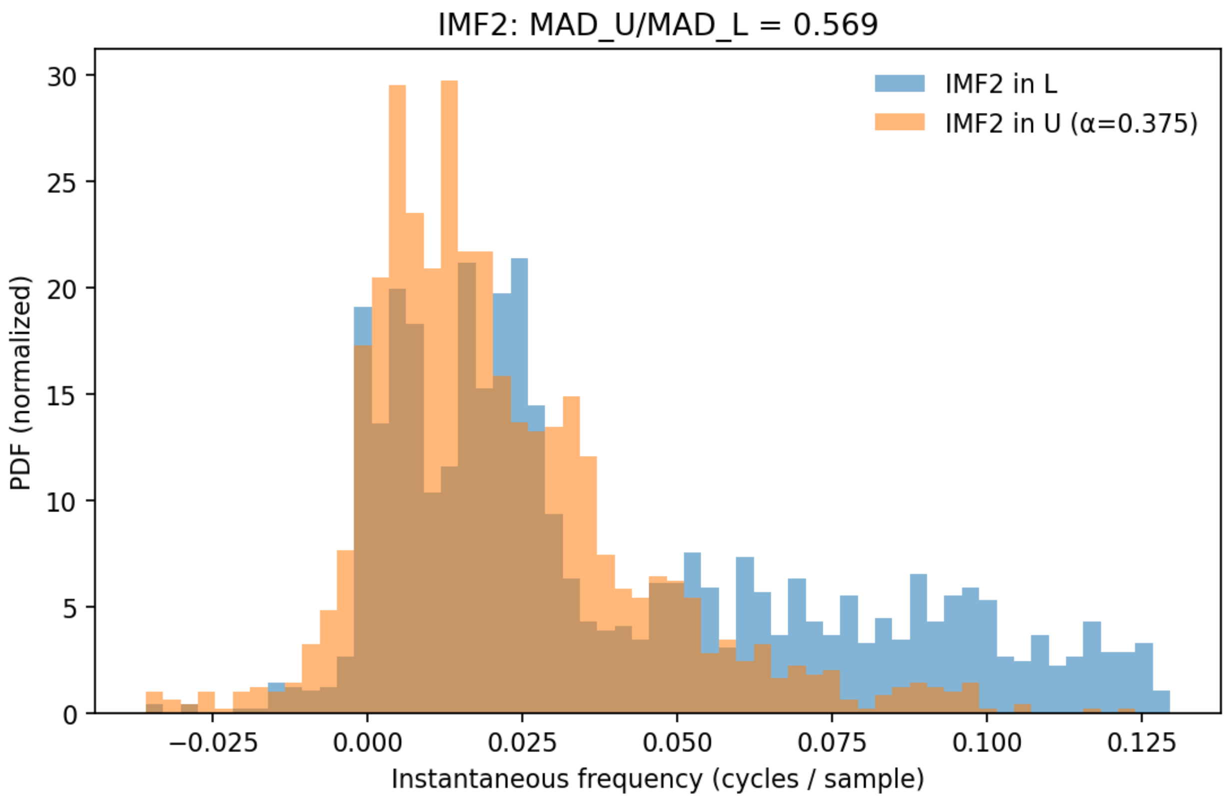

Fig. 1 — IMF2 instantaneous frequency histograms in L (blue) vs U (orange; α=0.375). The U-space distribution narrows; reported ratio MAD_U/MAD_L ≈ 0.569.

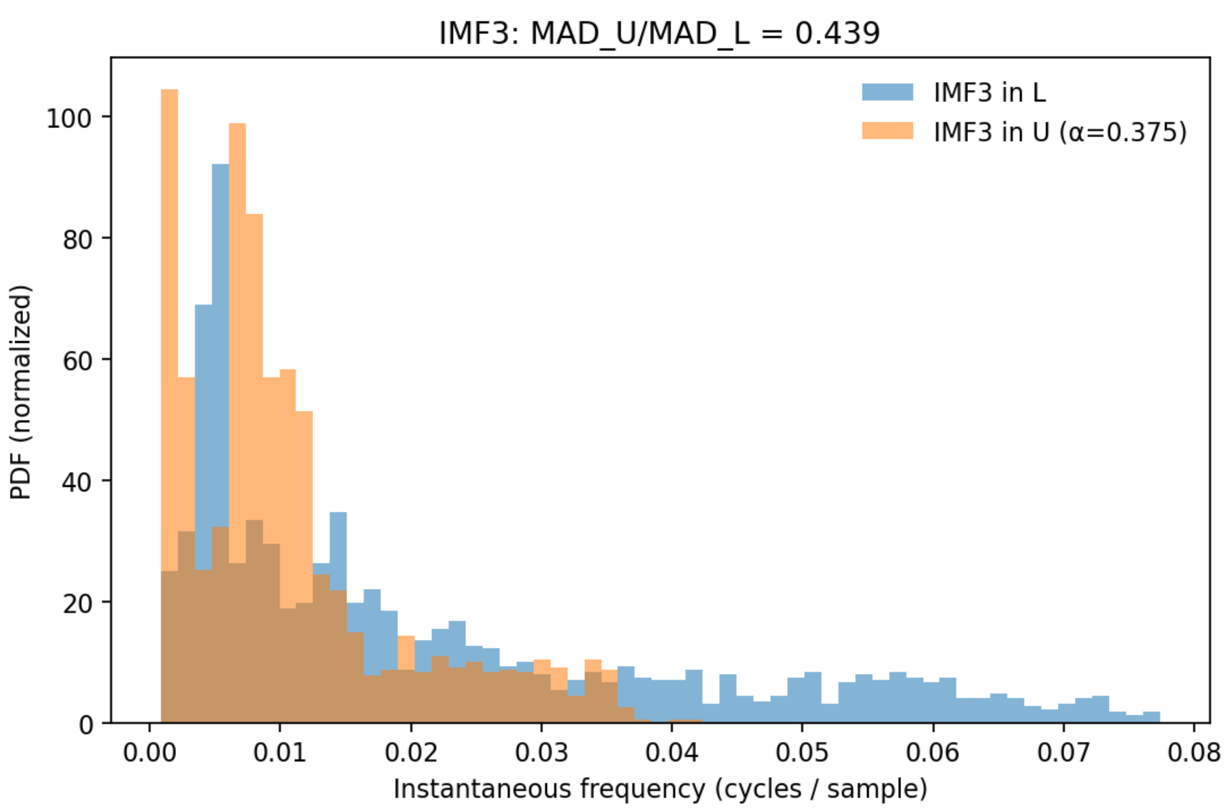

Fig. 2 — IMF3 instantaneous frequency histograms in L (blue) vs U (orange; α=0.375). Pronounced tightening; MAD_U/MAD_L ≈ 0.439.

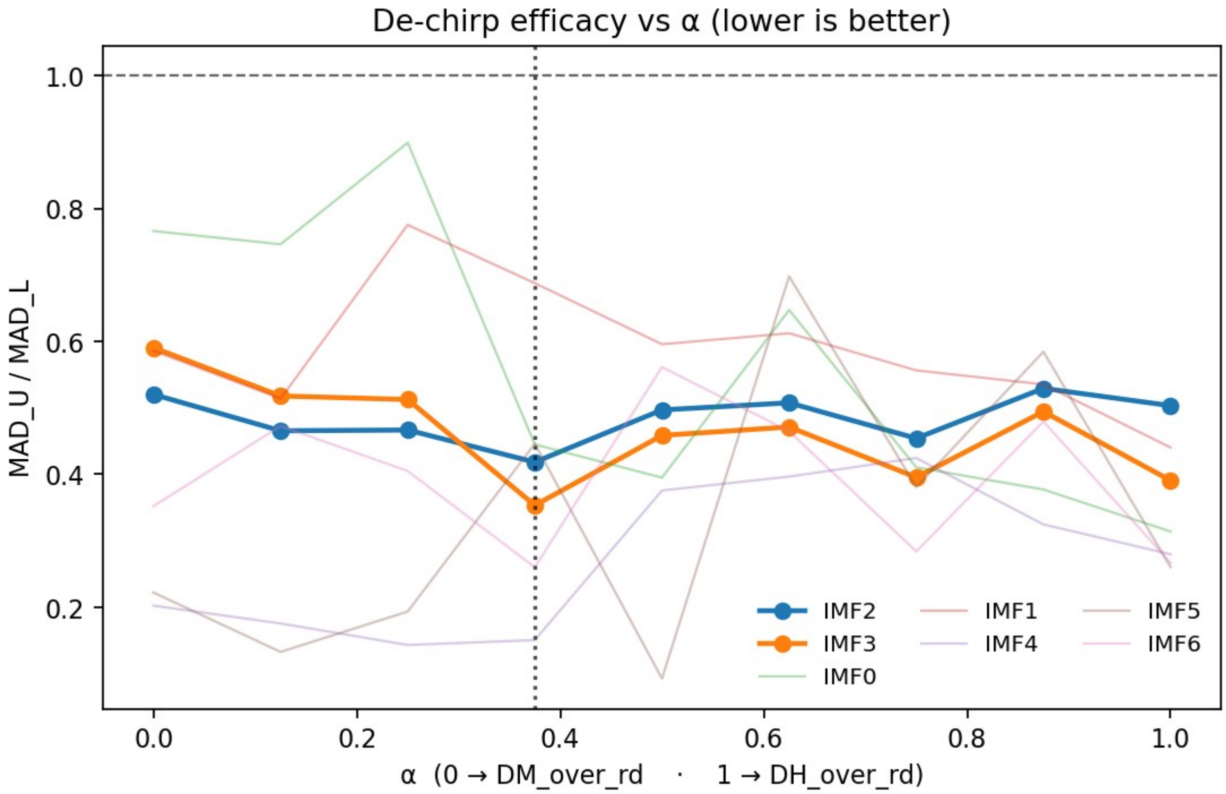

Fig. 3 — α sweep of MAD_U/MAD_L. IMF2–3 achieve their minimum near α≈0.375 (vertical line).

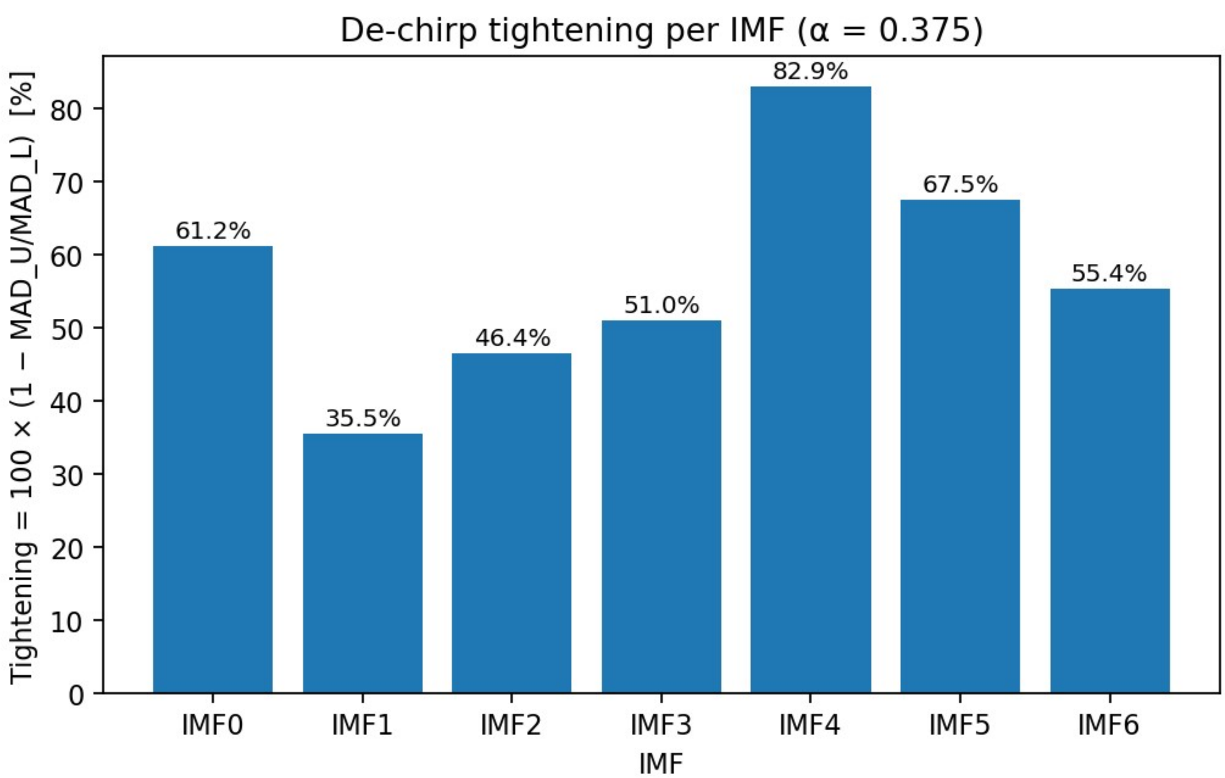

Fig. 4 — Percent tightening across IMFs at α=0.375. IMF4 shows the largest tightening, with IMF2–3 also significantly reduced.

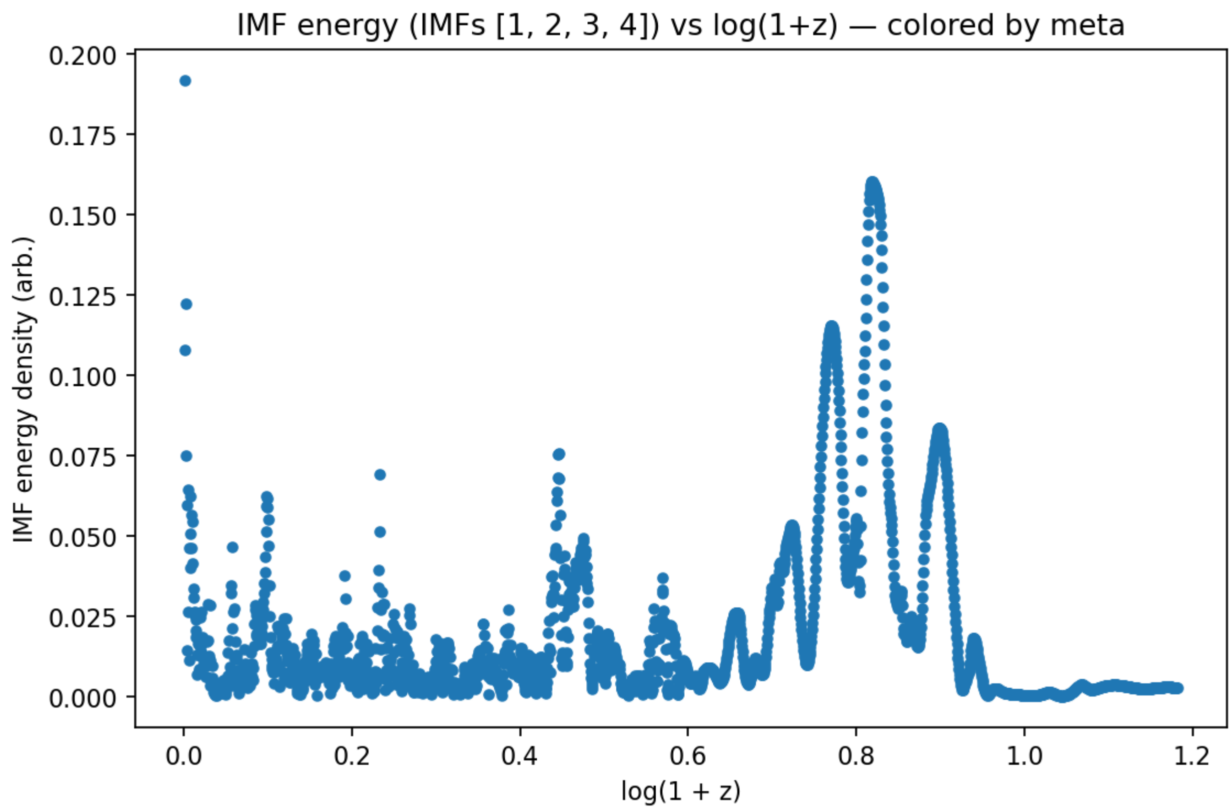

Fig. 5 — IMF energy density (sum over focus IMFs) vs L. Peaks indicate localized energy concentrations in log(1+z).

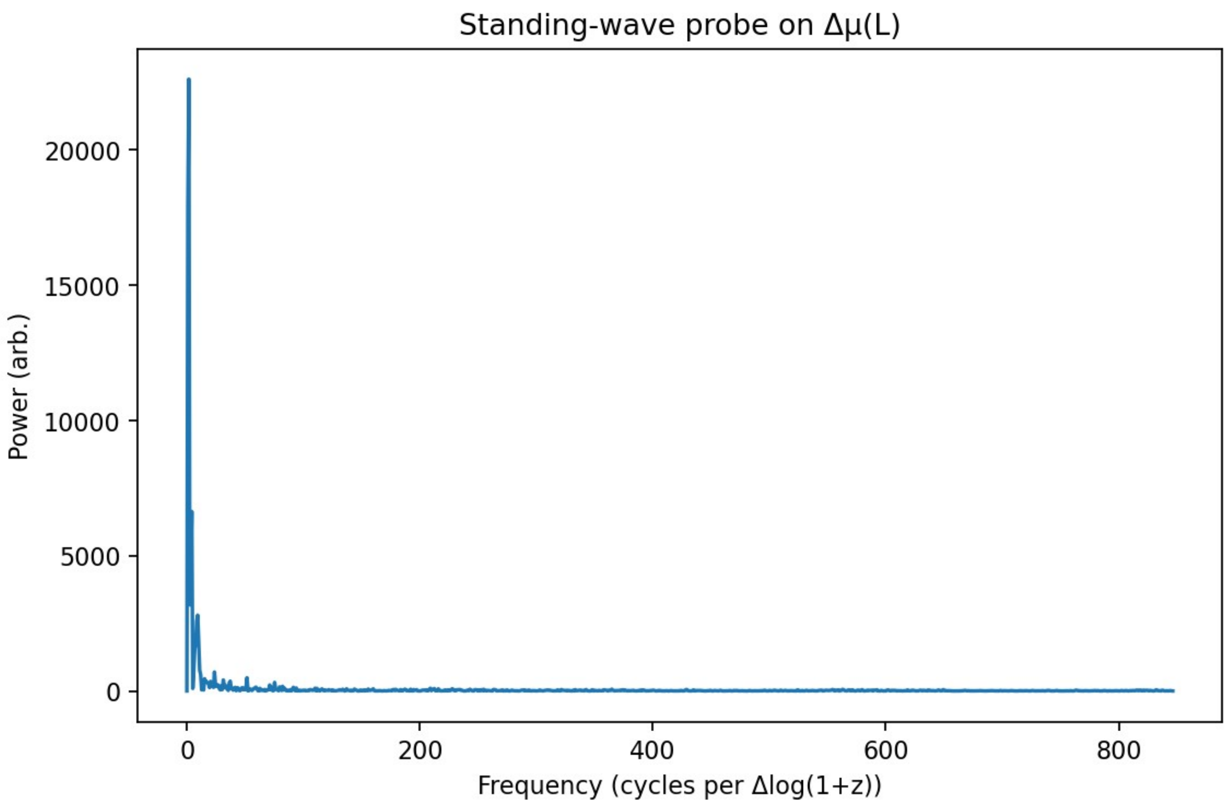

Fig. 6 — Standing-wave probe on Δμ(L). The near-zero-frequency dominance is consistent with a chirp rather than a strong stationary line.

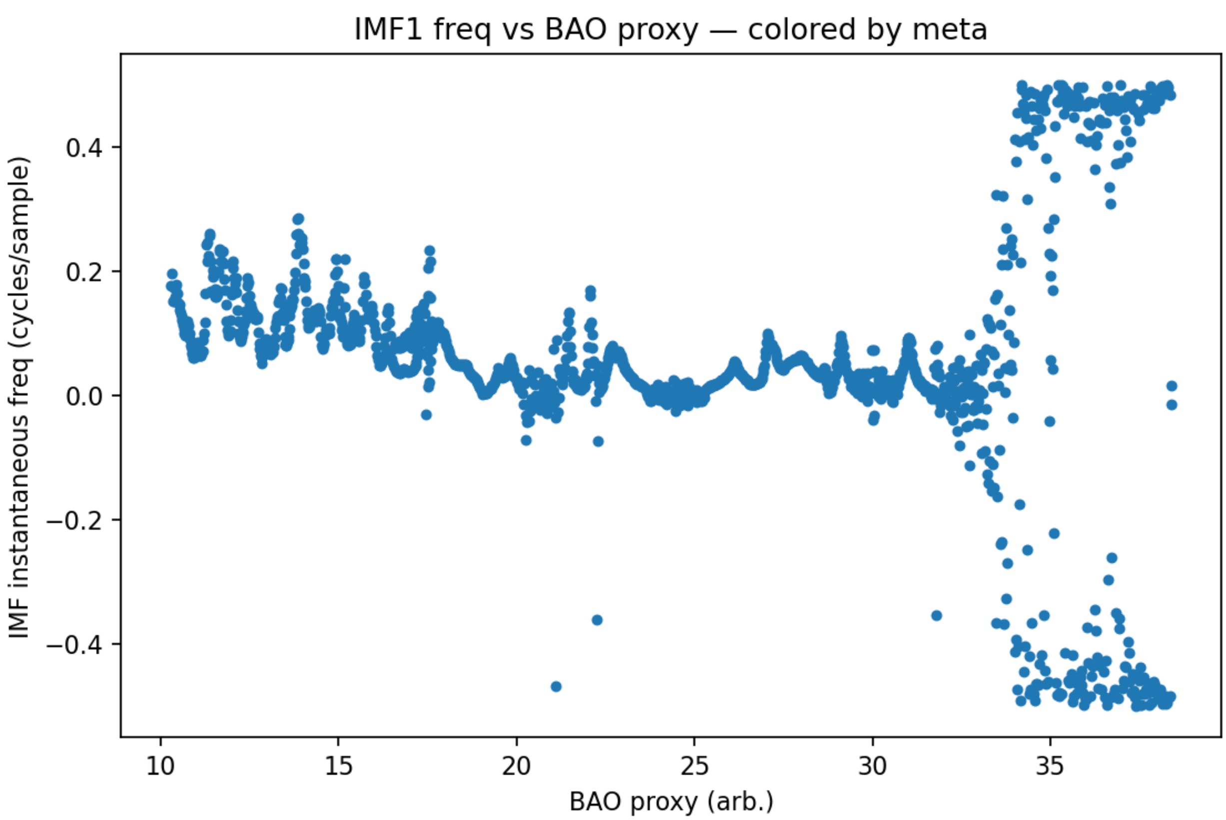

Fig. 7 — IMF1 instantaneous frequency vs BAO proxy on the L grid.

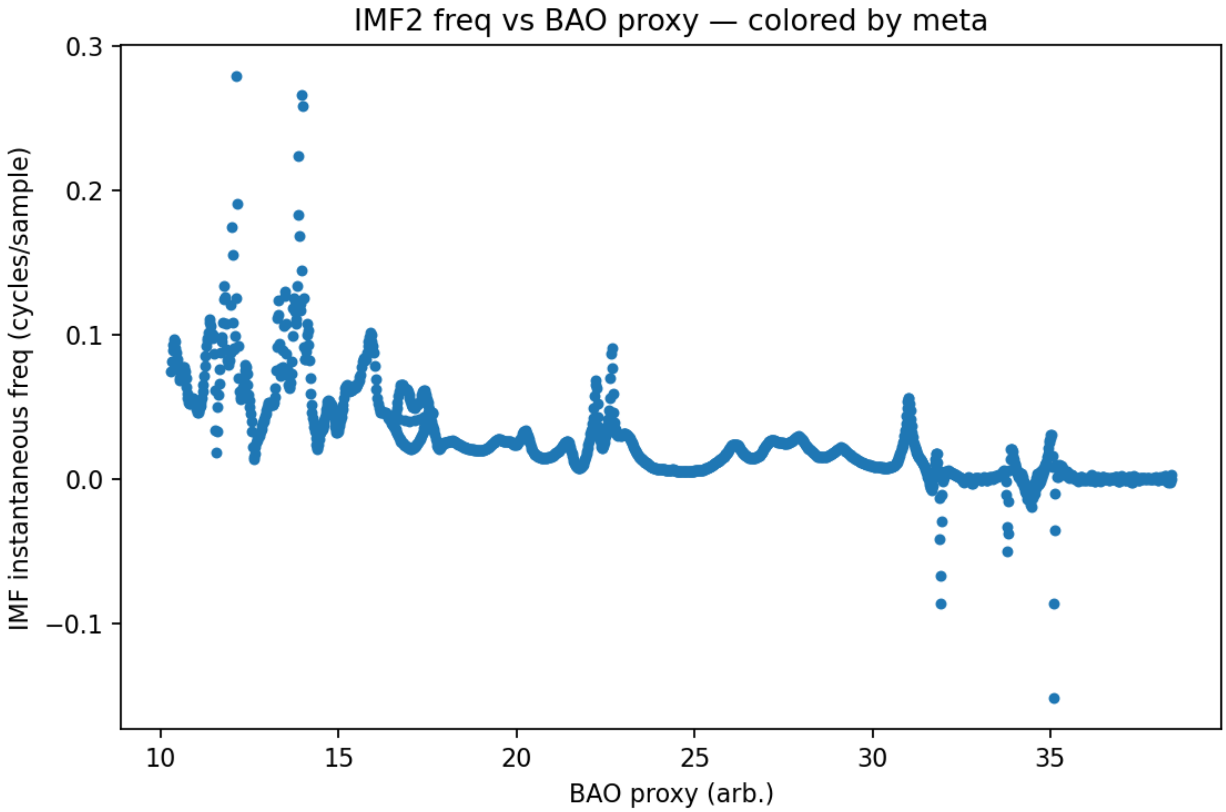

Fig. 8 — IMF2 instantaneous frequency vs BAO proxy on the L grid.

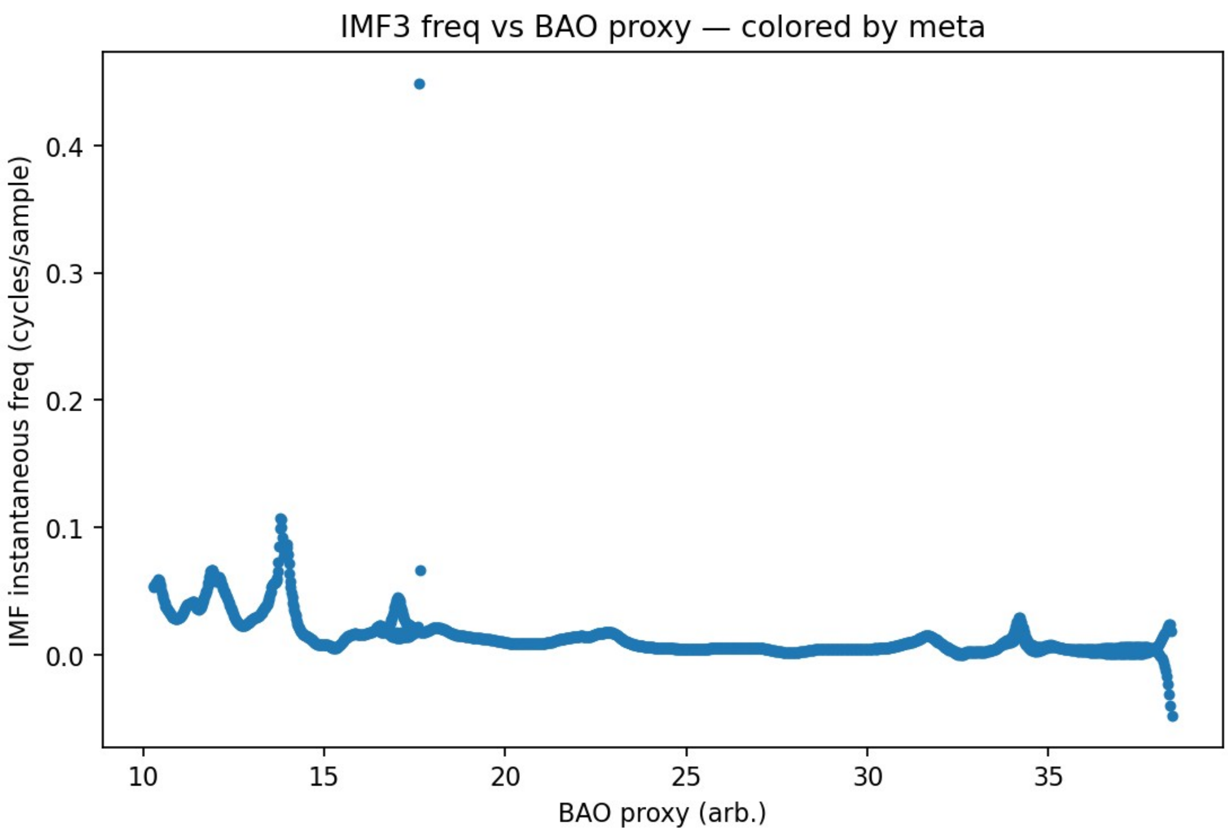

Fig. 9 — IMF3 instantaneous frequency vs BAO proxy on the L grid.

WL sweep for locking metric

WL	stat_obs	p_value	IMF1	IMF2	IMF3	IMF4
0.030	48.970	0.000	-0.050913	-0.068552	-0.048556	0.061210
0.035	48.107	0.000	-0.044921	-0.071515	-0.066169	0.076311
0.040	50.623	0.000	-0.039372	-0.083078	-0.090185	-0.072064
0.045	50.787	0.000	-0.045422	-0.105	-0.111	-0.065604
0.050	50.439	0.000	-0.053482	-0.115	-0.130	-0.025598

Lock scores (signed correlation-like statistic) become more negative for IMF2–3 with increasing WL, supporting a robust anti-correlation between IMF frequency and the BAO proxy when averaged over a broader L range. By a simple |IMF2,IMF3| figure-of-merit, WL≈0.050 is optimal in this sweep. All runs show p≈10^{-4} under phase-randomized surrogates.

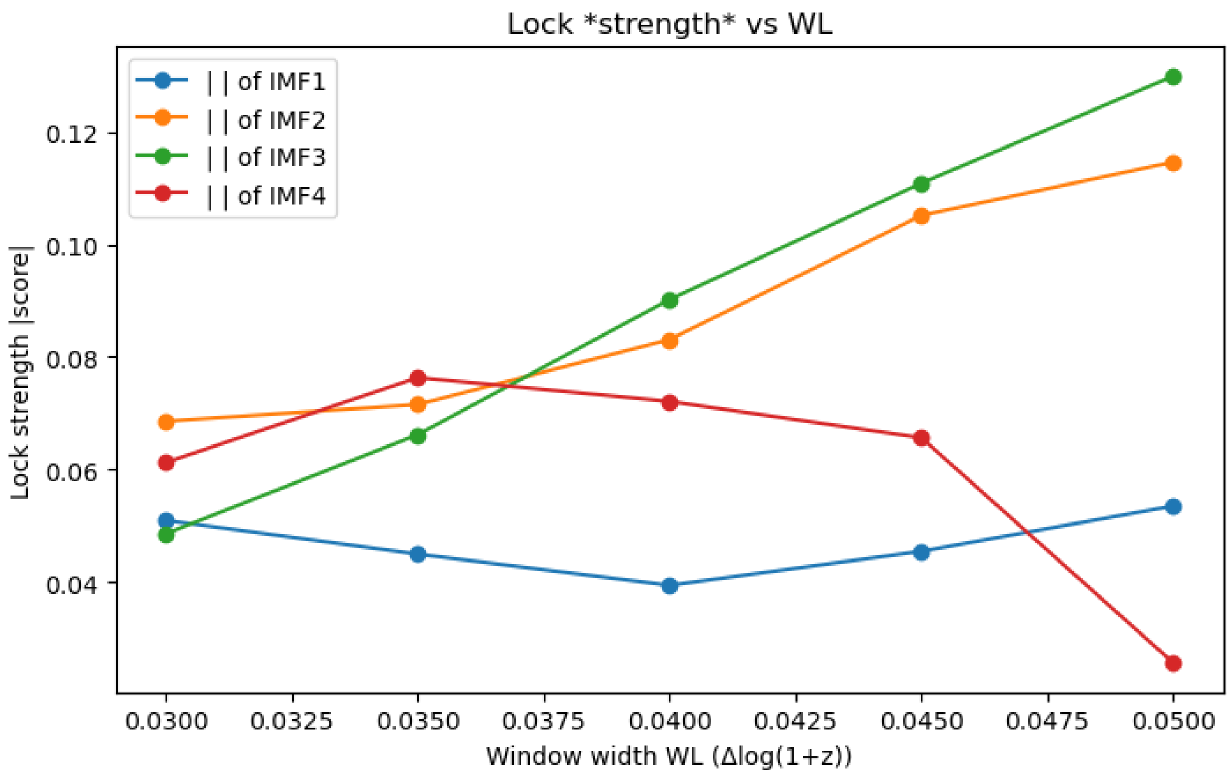

Fig. 10 — Lock *strength* (|score|) vs WL for IMFs 1–4.

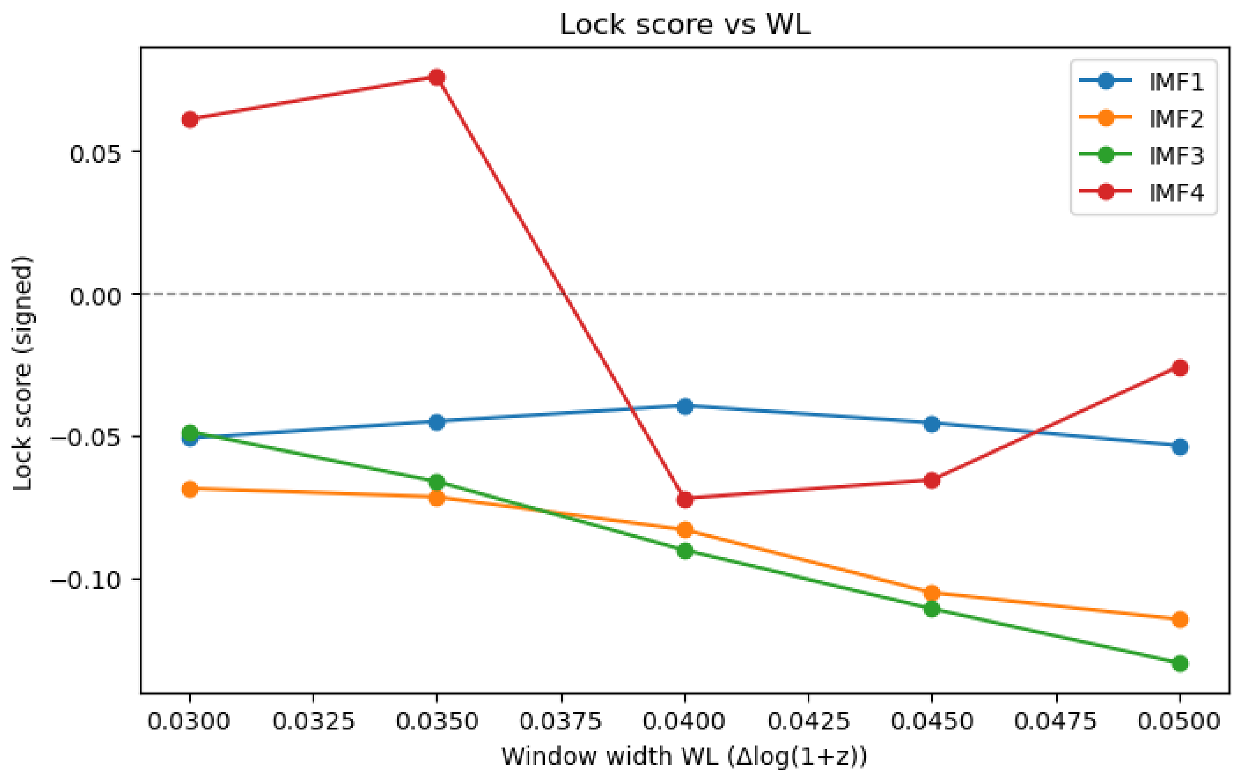

Fig. 11 — Signed lock score vs WL for IMFs 1–4.

Methods & scripts used

• imf23_probe.py (v0V1): loads SN/BAO, runs CEEMDAN, computes Hilbert analytics, lock metrics, surrogates, optional de-chirp in U.

• report_plots.py: builds publication figures (histograms L vs U, α-sweep, tightening bar).

• helpers_signal.py: interpolation utilities, de-chirp mapping helpers, and robust MAD/bootstrapping utilities used earlier.

• Notebooks/console harness: parameter sweeps (α, WL), dataset selection, and summary CSV generation.

Conclusion

The BAO-informed de-chirp U(L) substantially tightens IMF2–3 instantaneous-frequency distributions of SN Ia residuals. The $\alpha \approx 0.375$ optimum suggests a specific linear combination of transverse and radial BAO proxies that best linearizes the ridge. Consistently negative locking scores (stronger with larger WL) and $p \approx 10^{-4}$ surrogate significance indicate the effect is unlikely to arise from chance.

Conjecture

IMF2–3 in $\Delta\mu(L)$ likely encode a slowly varying, BAO-scale-modulated chirp. Constructing U from $b\alpha$ partially removes this modulation, revealing narrower, more stationary frequency content. If confirmed across independent SN/BAO catalogs, this would be consistent with a genuine SN–LSS coupling at BAO-effective scales rather than a processing artifact.